

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

23rd May 2023

**Subject SP7 – General Insurance Reserving and
Capital Modelling**

Time allowed: 3 Hours 15 Minutes (14.45 – 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 3. Attempt all questions beginning your answer to each question on a separate sheet.*
- 4. Mark allocations are shown in brackets.*
- 5. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** An insurance company XYZ is operating in a country where reserves are not allowed to be discounted by the Regulator for preparation of account. The Regulator has recently changed guidelines and allowed to discount only IBNR of long tailed class Motor TP due to significant delays in claim settlement.

Incurred development triangle with calculated Ultimate amounts are provided as on 31st March 2023. Calculate both undiscounted and discounted reserve at 5% p.a. State your assumptions.

Development of Incurred Triangle

Accident year / Dev year	1	2	3	4	5	Ultimate amount	Earned Premium
2018-2019	820000	2760000	3980000	4860000	5700000	6270000	6700000
2019-2020	780000	2420000	3760000	4830000		6279000	8200000
2020-2021	830000	2890000	4610000			6915000	10500000
2021-2022	1010000	3610000				8122500	11300000
2022-2023	1110000					8880000	11200000

[15]

- Q. 2)** Describe any two methods to separate reserves between IBNR and IBNER. Suggest challenges in this exercise.

[5]

- Q. 3)** A Company which only writes Crop Insurance has decided to enter Government mass health insurance scheme for growth in business. Company currently writes 5000 crores of crop insurance and plans to write further 5000 crores next year through Government schemes. There are various states in the country and each state has its own Health mass scheme which is written through Tender by State Health Agency (SHA). Insurance company which quotes lowest amount wins the Tender and gets the business. The company's current solvency is 1.75 and as per regulation minimum solvency requirement is 1.5.

Premium is received in 3 instalments from any SHA: 40% is paid at start of the policy, further 40% is paid after 6 months and remaining 20% after 10 months of policy start date. Policy period is one year which may be extended for a few months by SHA on pro rata basis of initial quoted premium.

As per terms and conditions of policy, after adjusting flat 7% percent of premium towards administrative cost (including all costs excluding only service tax and any cess, if applicable) and after settling of all claims, if there remains surplus then 100 percent of surplus should be refunded by the Insurer to the SHA.

Outline the various risks faced by the company and mitigation steps company can take to overcome these risks.

[20]

- Q. 4)** Insurer C is a General Insurance Company which started transacting household buildings and contents insurance on 1st January 2019.

- i) All business written in a year is based on the same premium rating basis. All reinsurance is on a quota share basis. You have been provided with the following accounting information:

				in (000 INR)
Underwriting Year	2019	2020	2021	2022
Net (of Reinsurance) Written Premium	25,000	50,000	100,000	200,000
Net Earned Premium	10,000	35,000	70,000	140,000
Net Loss Ratio (on occurrence basis)	150%	130%	100%	75%

Calculate gross loss ratio for each of the four underwriting years, stating any assumptions that you make. (8)

ii) Explain why the figures in (i) are unlikely to occur in practice. (5)

iii) Discuss the effects of rapid premium growth on an insurer i.e. balance sheet and future business plans. (5)

iv) Insurer (C) plans to introduce Engineering project policy on 1.1.2023. This risk has EML of INR 10,000,000. It has purchased excess of loss and surplus reinsurance for these policies.

a) Explain what is meant by EML. (2)

b) Write down the maximum EML that can be insured and not go beyond the maximum reinsurance cover, if the only reinsurance the company has available is a surplus treaty with N lines of cover and a maximum retention of R. (1)

c) C has a risk excess of loss contract INR 5,000,000 xs INR 1,000,000 with reinsurer A which applies before a surplus treaty with reinsurer B. The surplus treaty provides a maximum retention of INR 3,000,000 and 4 lines with the minimum always being ceded. Determine how much A, B and C each pay if a claim for INR 9,000,000 occurs. (4)
[25]

Q. 5) A reinsurance company wants to introduce a new type of aggregate excess of loss treaty covering subsidence claims on household insurance business. The treaty will cover all subsidence claims notified, during a calendar year, to an insurer. The treaty will be subject to an aggregate deductible and an aggregate limit.

i) State the characteristics of subsidence claims likely to be experienced by the insurer. (5)

ii) List the information likely to be required from other than the insurer in order to price this treaty. (5)
[10]

Q. 6)

i) State what is meant by a soft market and explain how it occurs and persists. (5)

ii) Describe the risks to a general insurance company of under-estimating the softness in the market. (10)
[15]

Q. 7) A general Insurance company only underwrites Motor business in India. State the factors which the company needs to consider while formulating its investment strategy. [10]
